AN UNRAMIFIED PRE-RELATION WITH POINTWISE EQUIVALENCE RELATION AND GROUPOID STRUCTURE MAPS IS AN EQUIVALENCE RELATION

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-unramified-permanence (canonical tag 02GG). cited at proof lines 444-446. j is unramified since s, t are.
- Lemma morphisms-lemma-diagonal-unramified-morphism (canonical tag 02GE). cited at proof lines 447-448. The diagonal Delta_j of a unramified morphism is an open immersion.

Source reference. Original lemma 02GG: sources/stacks-chow/tags.txt, source lines 3227–3227. Cited at proof lines 444-446: condition (1) makes j unramified; load-bearing.

Source reference. Original lemma 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Cited at proof lines 447-448: Delta_j is an open immersion; load-bearing.

Lemma 0.2. *Let $j : R \to U \times_S U$ be a pre-relation. Assume*

(1) *s, t are unramified,*
(2) *for any algebraically closed field k over S the map $R(k) \to U(k) \times U(k)$ is an equivalence relation,*
(3) *there are morphisms $e : U \to R$, $i : R \to R$, $c : R \times_{s,U,t} R \to R$ such that*

$$\begin{array}{ccc} U & \xrightarrow{e} & R \\ {\scriptstyle\Delta}\downarrow & & \downarrow{\scriptstyle j} \\ U \times_S U & \longrightarrow & U \times_S U \end{array} \qquad \begin{array}{ccc} R & \xrightarrow{i} & R \\ \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} \\ U \times_S U & \xrightarrow{flip} & U \times_S U \end{array} \qquad \begin{array}{ccc} R \times_{s,U,t} R & \xrightarrow{c} & R \\ \downarrow{\scriptstyle j\times j} & & \downarrow{\scriptstyle j} \\ U \times_S U \times_S U & \xrightarrow{pr_{02}} & U \times_S U \end{array}$$

are commutative.

Then j is an equivalence relation.

Proof. By condition (1) and Morphisms, Lemma 02GG we see that j is a unramified. Then $\Delta_j : R \to R \times_{U \times_S U} R$ is an open immersion by Morphisms, Lemma 02GE. However, then condition (2) says Δ_j is bijective on k-valued points, hence Δ_j is

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an isomorphism, hence j is a monomorphism. Then it easily follows from the commutative diagrams that $R(T) \subset U(T) \times U(T)$ is an equivalence relation for all schemes T over S. $\square$